Biological Molecules

Plan for the day

- Chemical bonds
- Molecules of life
  - Carbs
  - Lipids
  - Proteins
  - Nucleic acids

Biological Macromolecules

What Kinds of Molecules Characterize Living Things?

- Molecules that make up living organisms:
  - Carbohydrates
  - Lipids
  - Proteins
  - Nucleic acids
- Most are **polymers** of smaller molecules called **monomers**.

Carbohydrates

3.3 What Are the Chemical Structures and Functions of Carbohydrates?

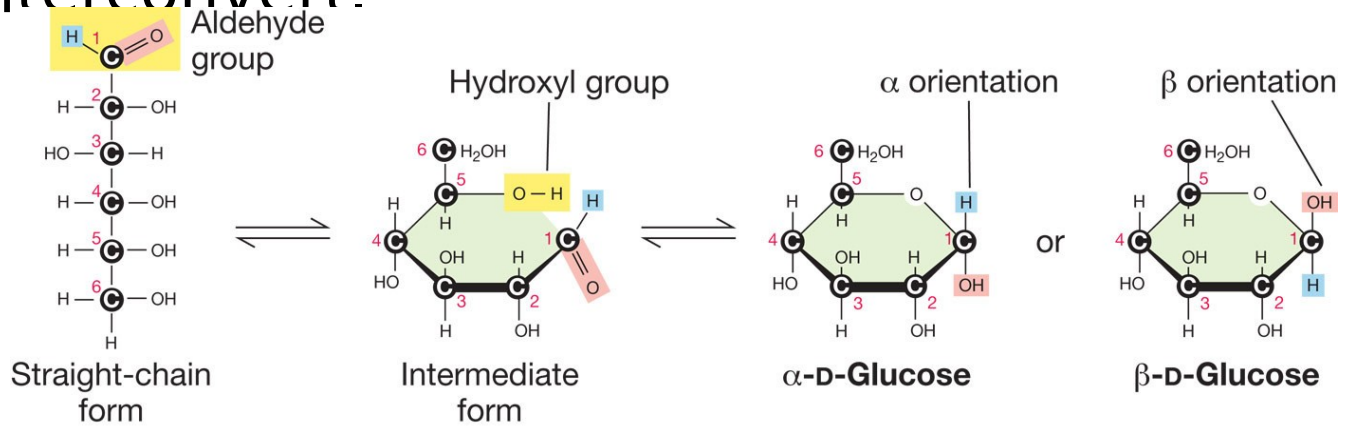
- **Carbohydrates** have the general formula $C_nH_{2n}O_n$. They are:
  - sources of stored energy
  - used to transport stored energy
  - carbon skeletons for many other molecules

3.3 What Are the Chemical Structures and Functions of Carbohydrates?

- **Monosaccharides**: simple sugars.
- **Disaccharides**: two simple sugars linked by covalent bonds.
- **Oligosaccharides**: 3 to 20 monosaccharides.
- **Polysaccharides**: hundreds or thousands of monosaccharides—e.g., starch, glycogen, cellulose.

3.3 What Are the Chemical Structures and Functions of Carbohydrates?

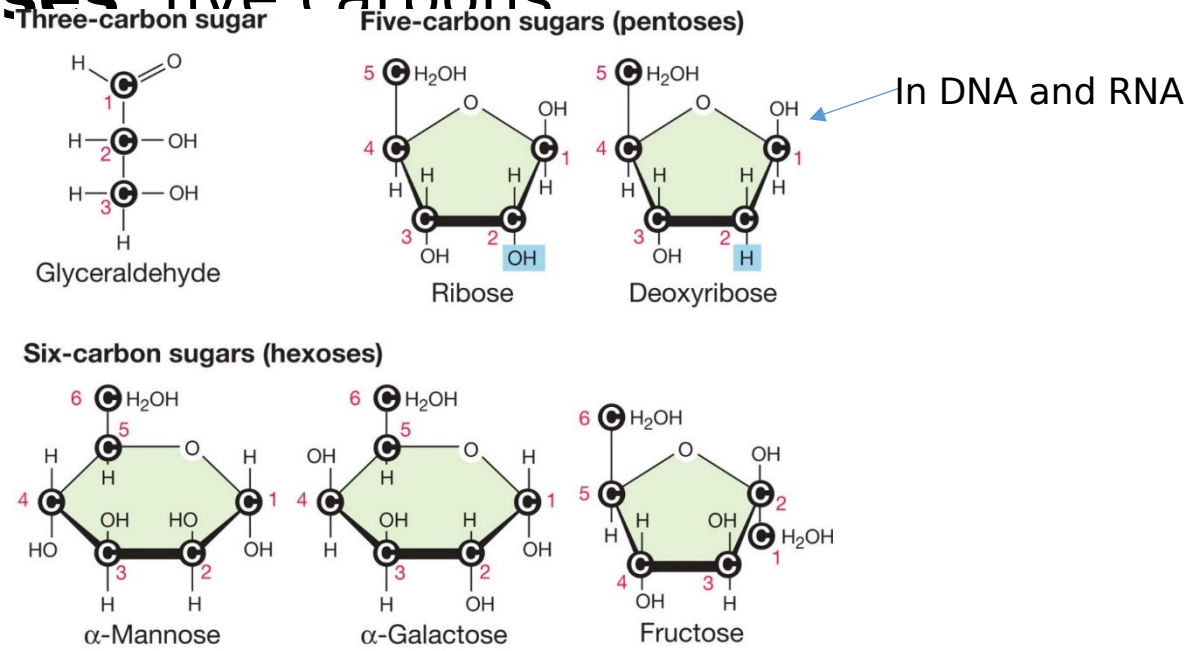
- All cells use **glucose** (monosaccharide) as an energy source.
- Exists as a straight chain or ring form. Ring is more common—it is more stable.
- Ring form exists as α- or β-glucose, which can interconvert.



LIFE 10e, Figure 3.15
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3.3 What Are the Chemical Structures and Functions of Carbohydrates?

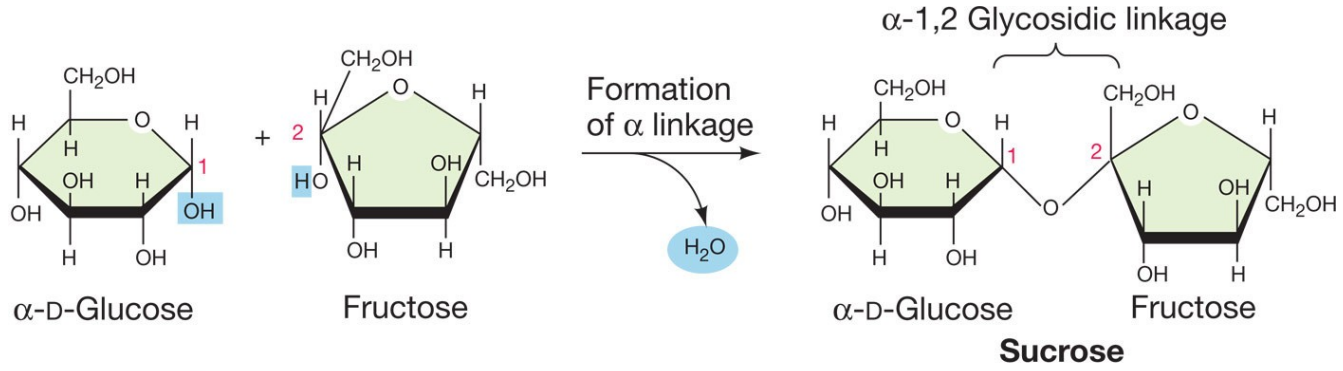
- Monosaccharides have different numbers of carbons:
- **Hexoses**: six carbons—structural isomers.
- **Pentoses**: five carbons



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3.3 What Are the Chemical Structures and Functions of Carbohydrates?

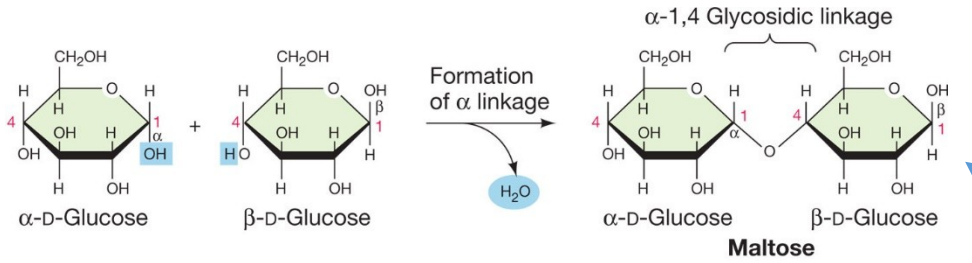
- Monosaccharides bind together in condensation reactions to form **glycosidic linkages**.
- Glycosidic linkages can be α or β.



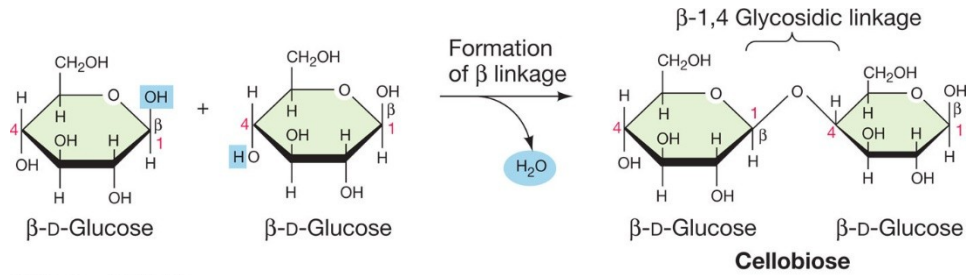
LIFE 10e, Figure 3.17 (Part 1)
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Figure 3.17 Disaccharides Form by Glycosidic Linkages (Part 2)

- Disachharides can vary based on the monosaccharide composition or the type of glycosidic linkage between them.



LIFE 10e, Figure 3.17 (Part 2)
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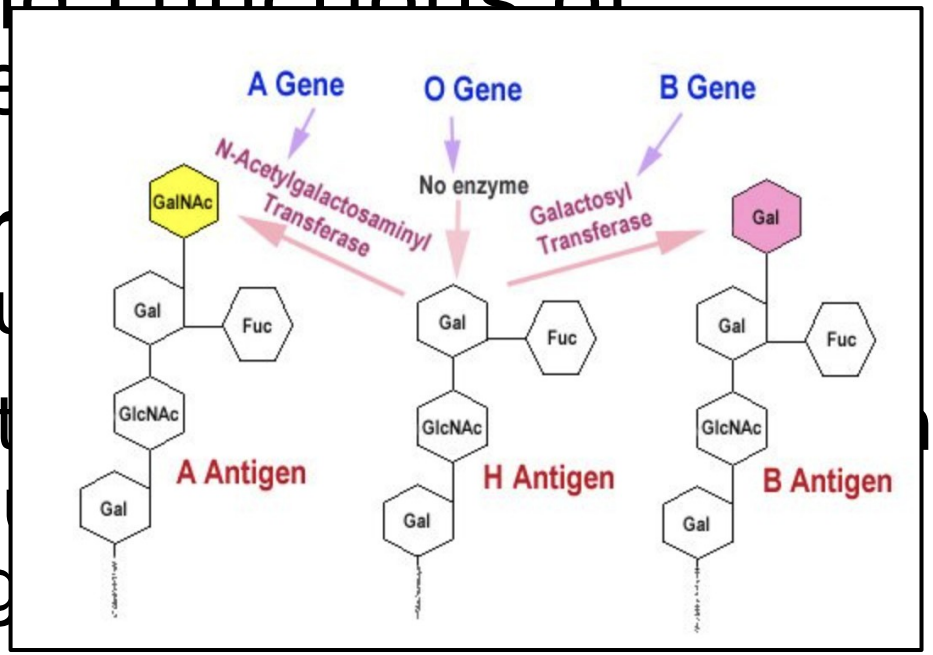
LIFE 10e, Figure 3.17 (Part 3)
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Same composition but different linkage characteristics

3.3 What Are the Chemical Structures and Functions of Carbohydrates?

- **Oligosaccharides** may include other functional groups.
- Often covalently bonded to proteins and lipids on cell surfaces and act as recognition signals. (virus capsid components)
- Human blood groups get specificity from oligosaccharide chains. These oligosaccharides are on the cell membrane, attached to proteins: Glycoproteins.

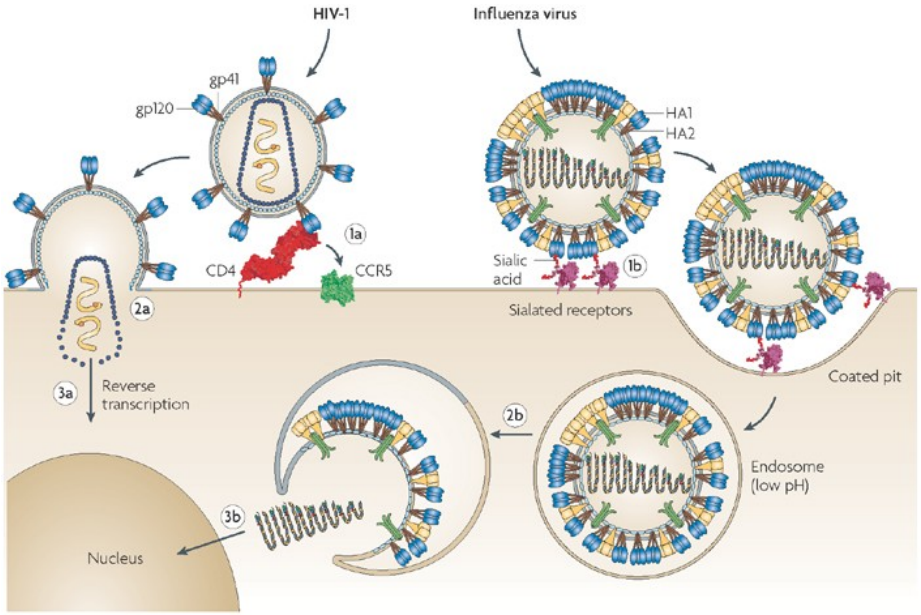
3.3 What Are the Chemical Structures and Functions of Carbohydrate



- **Oligosacchar**
  functional grou
- Often covalent d
  lipids on cell s
  recognition sig
  components)
- Human blood groups get specificity from oligosaccharide chains. These oligosaccharides are on the cell membrane, attached to proteins: Glycoproteins.

Virus entry

- Glycoproteins on the cell membrane distinguish target cells for viral entry.
- Virus glycoproteins recognize target ones and gain access to the cell.



https://www.researchgate.net/publication/5654699_The_challenge_of_eliciting_neutralizing_antibodies_to_HIV-1_and_to_influ

3.3 What Are the Chemical Structures and Functions of Carbohydrates?

- **Polysaccharides** are giant polymers of monosaccharides.
- **Starch**: storage of glucose in plants.
- **Glycogen**: storage of glucose in animals.
- **Cellulose**: very stable, good for structural components.

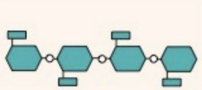
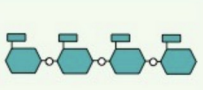
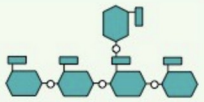
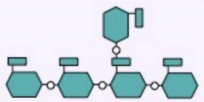
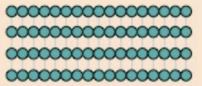
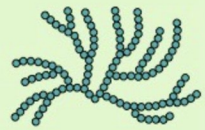
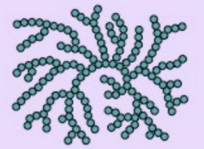
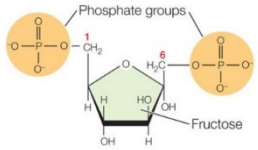
	Cellulose	Starch		Glycogen
		Amylose	Amylopectin	
Source	Plant	Plant	Plant	Animal
Subunit	β-glucose	α-glucose	α-glucose	α-glucose
Bonds	1-4	1-4	1-4 *and* 1-6	1-4 *and* 1-6
Branches	No	No	Yes (~per 20 subunits)	Yes (~per 10 subunits)
Diagram				
Shape				

Figure 3.19 Chemically Modified Carbohydrates

(A) Sugar phosphate

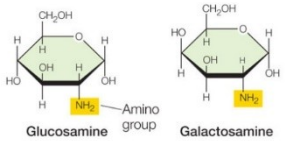
Fructose 1,6-bisphosphate is involved in the reactions that liberate energy from glucose. (The numbers in its name refer to the carbon sites of phosphate bonding; *bis-* indicates that two phosphates are present.)



Fructose 1,6-bisphosphate

(B) Amino sugars

The monosaccharides glucosamine and galactosamine are amino sugars with an amino group in place of a hydroxyl group.



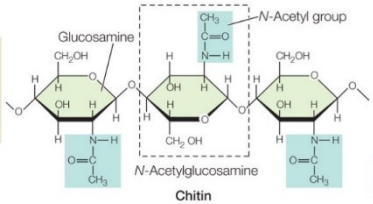
Glucosamine Galactosamine



Galactosamine is an important component of cartilage, a connective tissue in vertebrates.

(C) Chitin

Chitin is a polymer of *N*-acetylglucosamine; *N*-acetyl groups provide additional sites for hydrogen bonding between the polymers.



Chitin



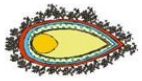
The external skeletons of insects are made up of chitin.

LIFE 10e, Figure 3.19
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Applications of chitin and chitosan

Application area	Specific use
Water treatment	Coagulating/flocculating agents for polluted waste waters
	Removal/recovery of metal ions from aqueous waste water
Agriculture	Plant elicitor
	Antimicrobial agents
	Plant seed coating
	Fertilizer
Textile and paper	Fibers for textile and woven fabrics
	Paper and film
Biotechnology	Chromatography packing
	Enzyme immobilizing materials
Food/Health Supplements	Natural thickeners
	Food additives including pet food
	Food processing (e.g. in sugar refining)
	Filtration and clarification
	Hypocholesterolemic agent (slimming agents)
Cosmetic	Ingredients for hair and skin care (conditioners)
Biomedical	Burns and wounds dressings for humans and animals
	Biomaterial (e.g. absorbable sutures)
	Anticoagulant or antithrombogenic materials (as sulfated-chitin derivatives)
	Hemostatic agents (as chitosan)
	Drug delivery, gene delivery

Table. Applications of chitin, chitosan and their derivatives [1]



Fungicide seed coating





Crini G. *Progress in Polymer Science*, 30 (2005) 38



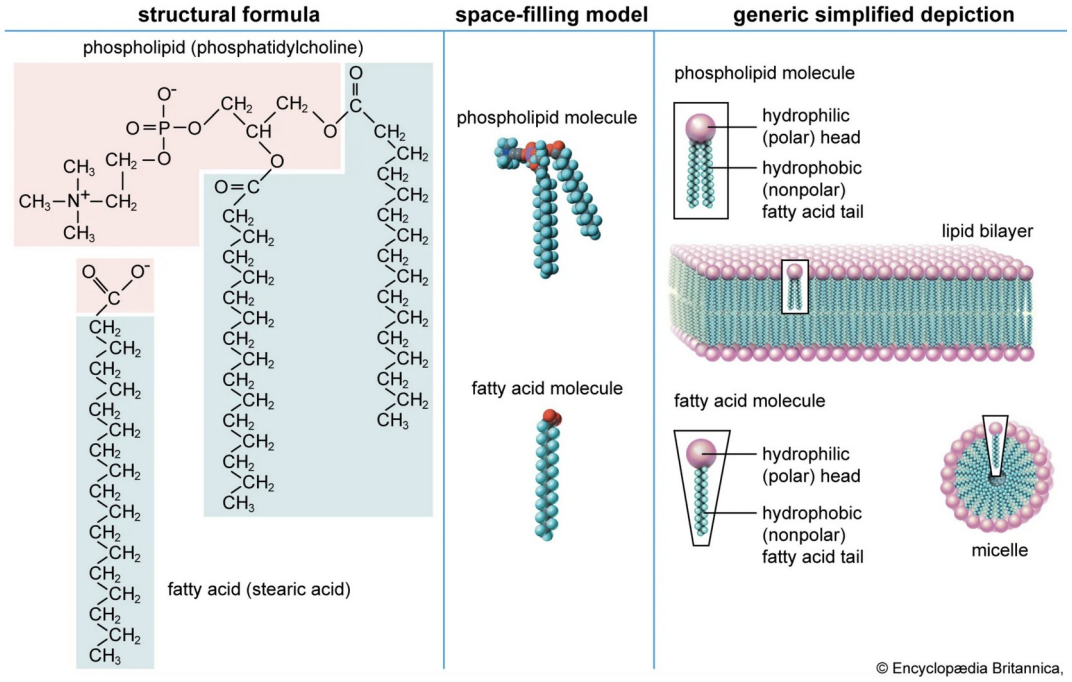
ChitoSan® fibers and chitin socks



Chitosan soap, lotion, shampoo

[1] Goosen M. *in Applications of Chitin and Chitosan*. Technomic Publishing Inc, PA. 1997

Lipids

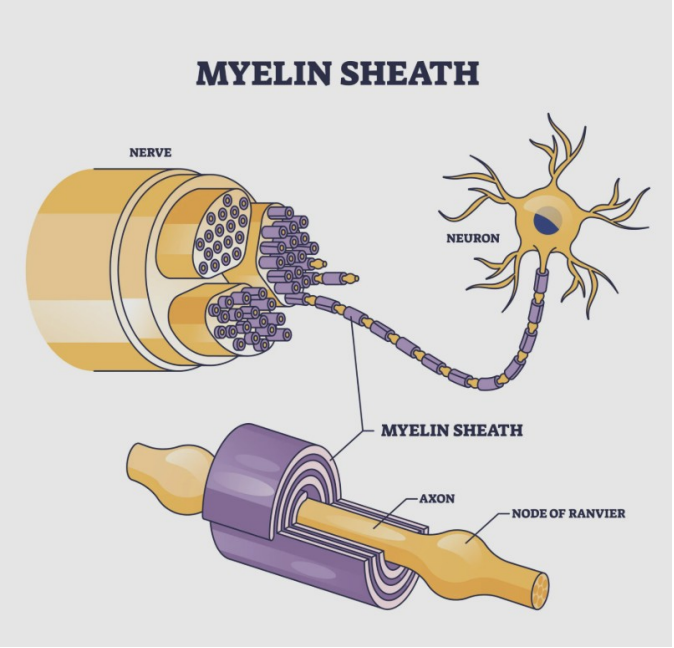

structural formula
space-filling model
generic simplified depiction
phospholipid (phosphatidylcholine)
fatty acid (stearic acid)
phospholipid molecule
fatty acid molecule
phospholipid molecule
hydrophilic (polar) head
hydrophobic (nonpolar) fatty acid tail
lipid bilayer
fatty acid molecule
hydrophilic (polar) head
hydrophobic (nonpolar) fatty acid tail
micelle
© Encyclopædia Britannica,

3.4 What Are the Chemical Structures and Functions of Lipids?

- **Lipids** are nonpolar hydrocarbons; insoluble in water.
- If close together, weak but additive van der Waals forces hold them together.
- Fats and oils store energy.
- Phospholipids—structural role in cell membranes.
- Carotenoids and chlorophylls—capture light energy in plants.
- Steroids and modified fatty acids—hormones and vitamins.

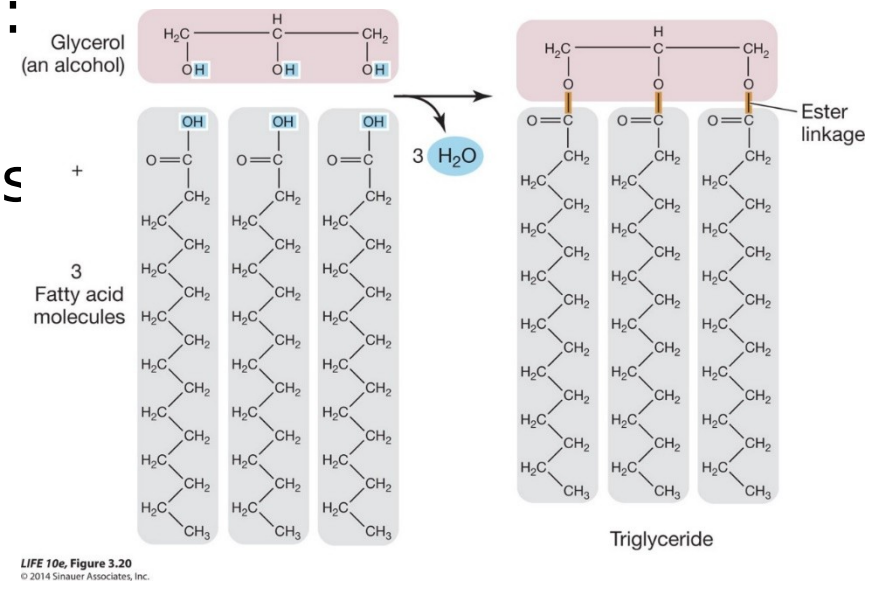
3.4 What Are the Chemical Structures and Functions of Lipids?

- Animal fat—thermal insulation.
- Lipid coating around nerves provides electrical insulation.
- Oil and wax on skin, fur, and feathers repels water.



3.4 What Are the Chemical Structures and Functions of Lipids?

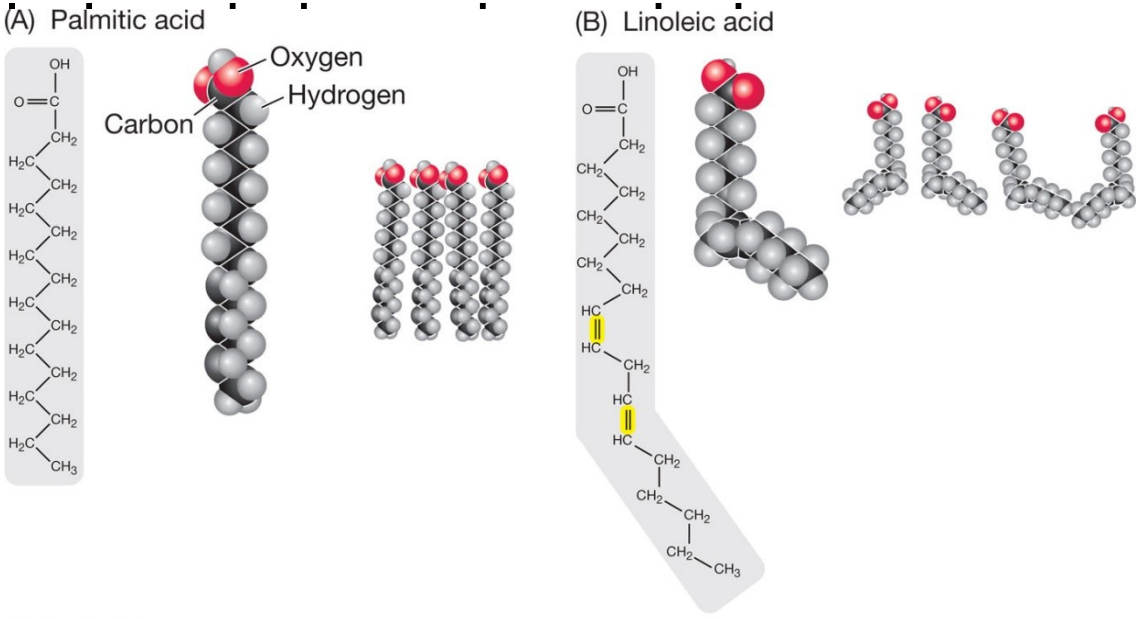
- Fats and oils are **triglycerides**: three fatty acids plus glycerol.
- **Glycerol**: has three –OH groups (an alcohol).
- **Fatty acid**: nonpolar hydrocarbon (CH) with a polar carboxyl group.
- Carboxyls bond with hydroxyls of glycerol in an **ester linkage**.



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3.4 What Are the Chemical Structures and Functions of Lipids?

- **Saturated fatty acid**: no double bonds between carbons—it is saturated with H atoms.
- **Unsaturated fatty acid**: one or more doub





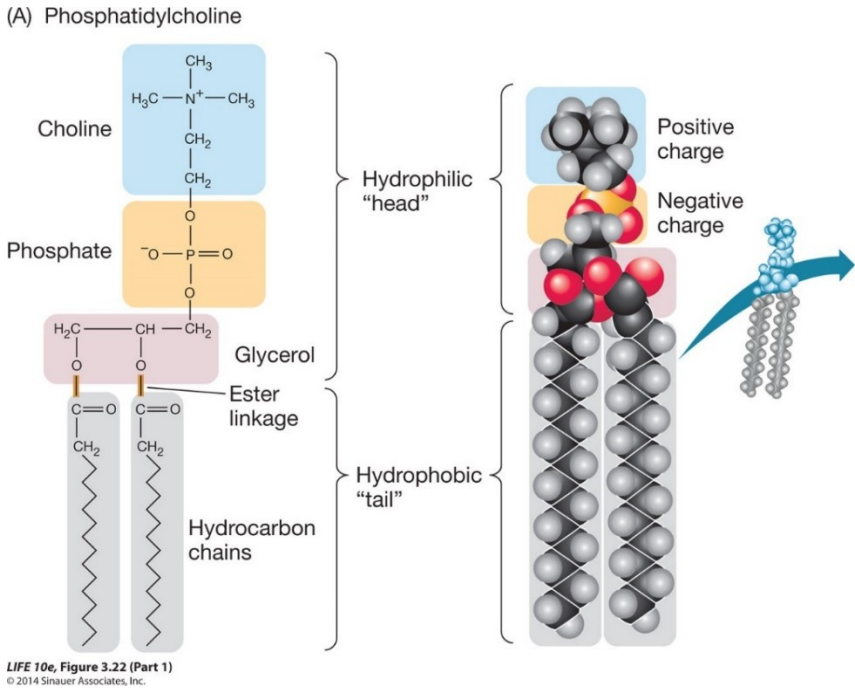
LIFE 10e, Figure 3.21
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3.4 What Are the Chemical Structures and Functions of Lipids?

- **Phospholipids**: fatty acids bound to glycerol; a phosphate group replaces one fatty acid.
    - The “head” is a phosphate group— hydrophilic
    - “Tails” are fatty acid chains—hydrophobic
    - They are amphipathic

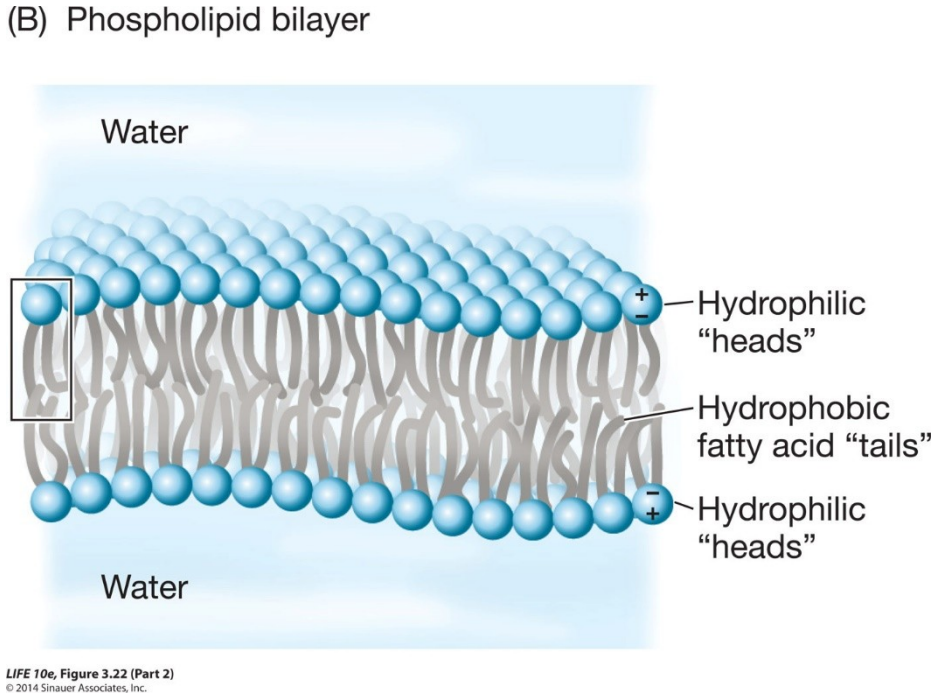
The head group of phospholipids can be varied: choline, inositol, serine, ethanolamine
These differences change the hydrophilicity and the size of the phospholipid



LIFE 10e, Figure 3.22 (Part 1)
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Figure 3.22 Phospholipids (Part 2)

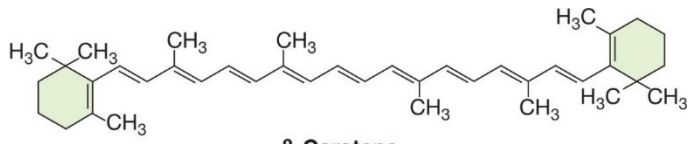
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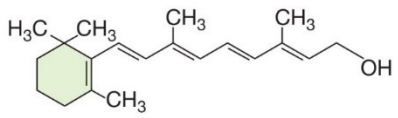
LIFE 10e, Figure 3.22 (Part 2)
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3.4 What Are the Chemical Structures and Functions of Lipids?

- **Carotenoids**: light-absorbing pigments
- **Vitamins**—small molecules not synthesized by the body; must be acquired in the diet.



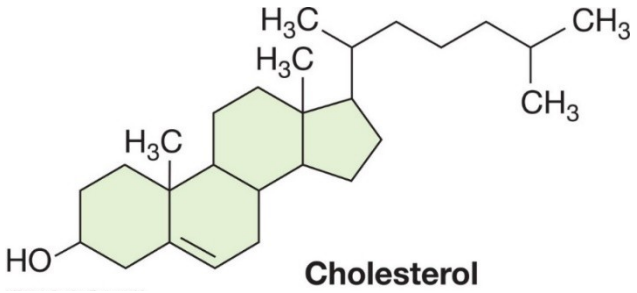
β-Carotene



Vitamin A

3.4 What Are the Chemical Structures and Functions of Lipids?

- **Steroids**: multiple rings share carbons.
- Cholesterol is important steroid in membranes; other steroids function as hormones.



Cholesterol

- **Waxes**—highly nonpolar and impermeable to water.

Proteins

Functions of Proteins

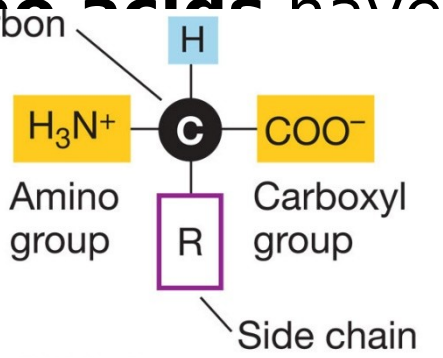
TABLE 3.1
Proteins and Their Functions

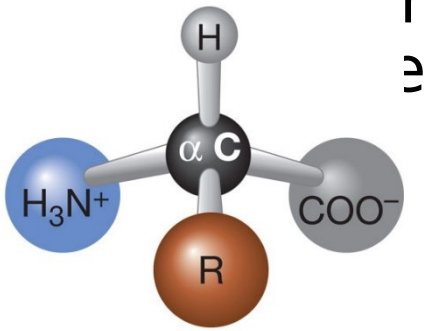
Category	Function
Enzymes	Catalyze (speed up) biochemical reactions
Structural proteins	Provide physical stability and movement
Defensive proteins	Recognize and respond to nonself substances (e.g., antibodies)
Signaling proteins	Control physiological processes (e.g., hormones)
Receptor proteins	Receive and respond to chemical signals
Membrane transporters	Regulate passage of substances across cellular membranes
Storage proteins	Store amino acids for later use
Transport proteins	Bind and carry substances within the organism
Gene regulatory proteins	Determine the rate of expression of a gene

LIFE 10e, Table 3.1
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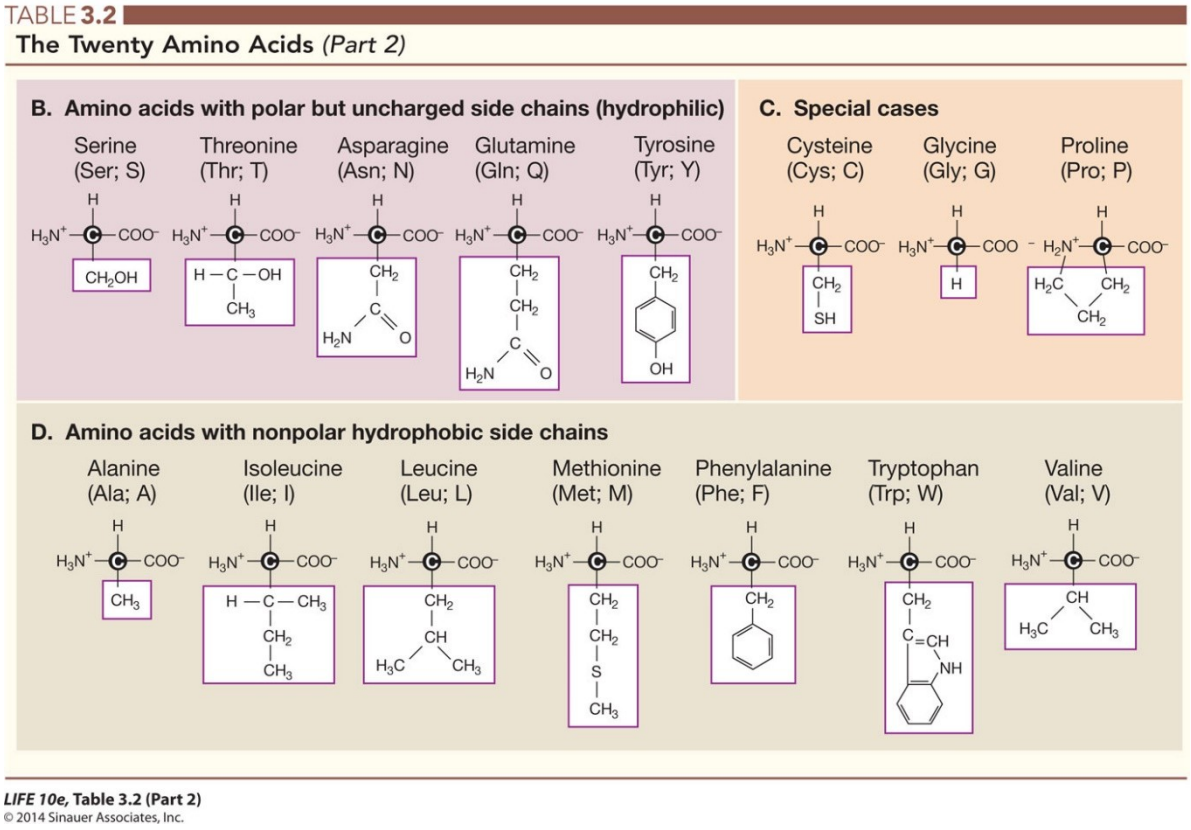
Chemical Structure of Proteins

- **Proteins** are polymers of 20 different amino acids.
- **Polypeptide chain**: single, unbranched chain of amino acids.
- Proteins consist of one or more polypeptide chains, which are folded into specific 3-D shapes defined by the sequence of amino acids.
- **Am**ino groups —t e.

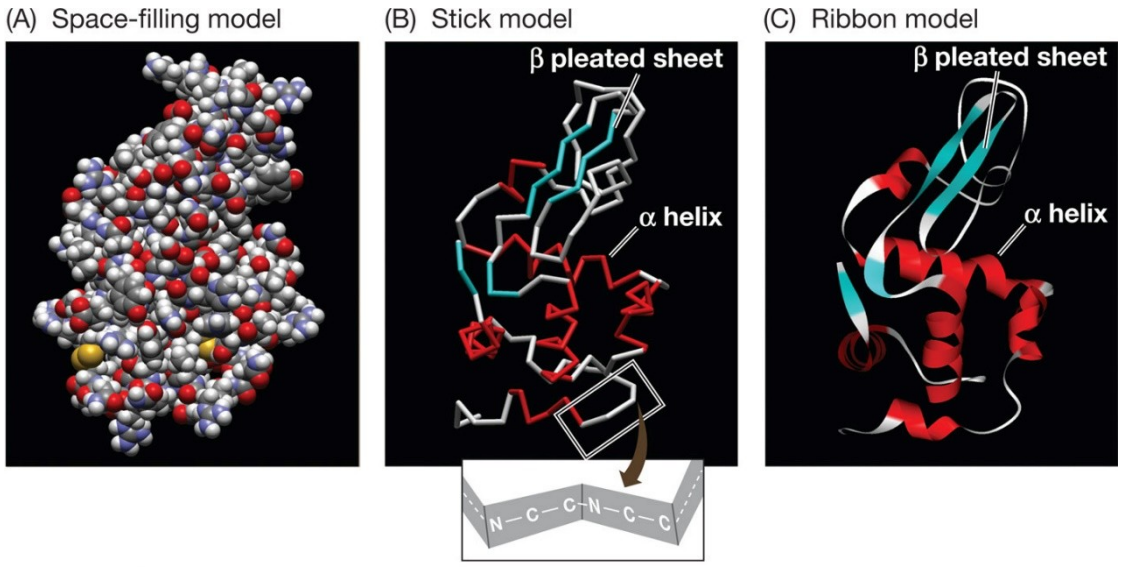




Aminoacids


TABLE 3.2
The Twenty Amino Acids (Part 2)
B. Amino acids with polar but uncharged side chains (hydrophilic)
Serine (Ser; S)
Threonine (Thr; T)
Asparagine (Asn; N)
Glutamine (Gln; Q)
Tyrosine (Tyr; Y)
C. Special cases
Cysteine (Cys; C)
Glycine (Gly; G)
Proline (Pro; P)
D. Amino acids with nonpolar hydrophobic side chains
Alanine (Ala; A)
Isoleucine (Ile; I)
Leucine (Leu; L)
Methionine (Met; M)
Phenylalanine (Phe; F)
Tryptophan (Trp; W)
Valine (Val; V)
LIFE 10e, Table 3.2 (Part 2)
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Protein structure can be represented in different ways (Lysozyme enzyme below)



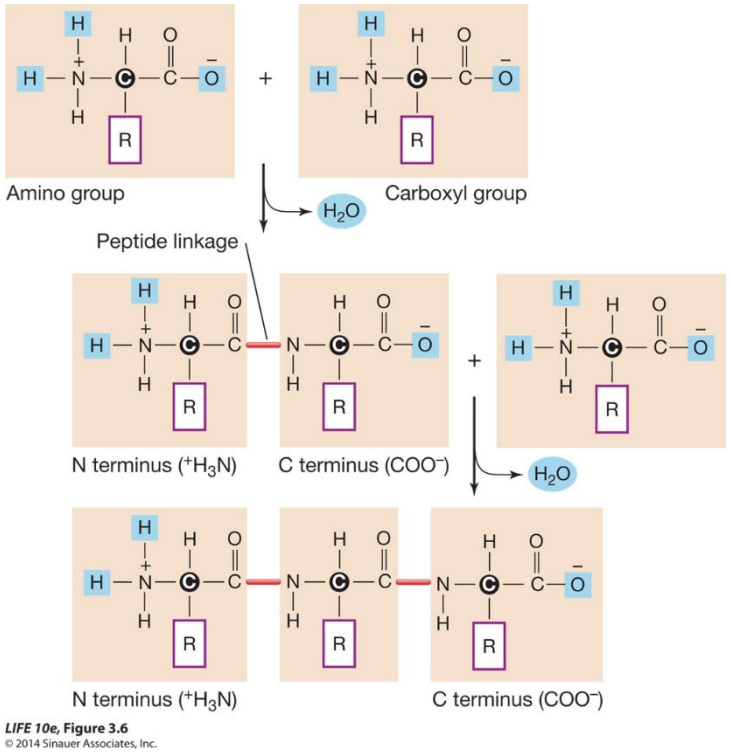
LIFE 10e, Figure 3.9
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How to fold a protein--

There are 4 levels of structure to achieve a three-dimensional protein structure.

1. Primary structure: aminoacid sequence
2. Secondary structure: alpha helices and beta sheets
3. Tertiary structure: Folding of secondary structure on itself
4. Quaternary structure: Some proteins are made from multiple polypeptide chains (primary sequence). Folding of independent polypeptide chains together is quaternary structure.

Formation of Peptide Linkages



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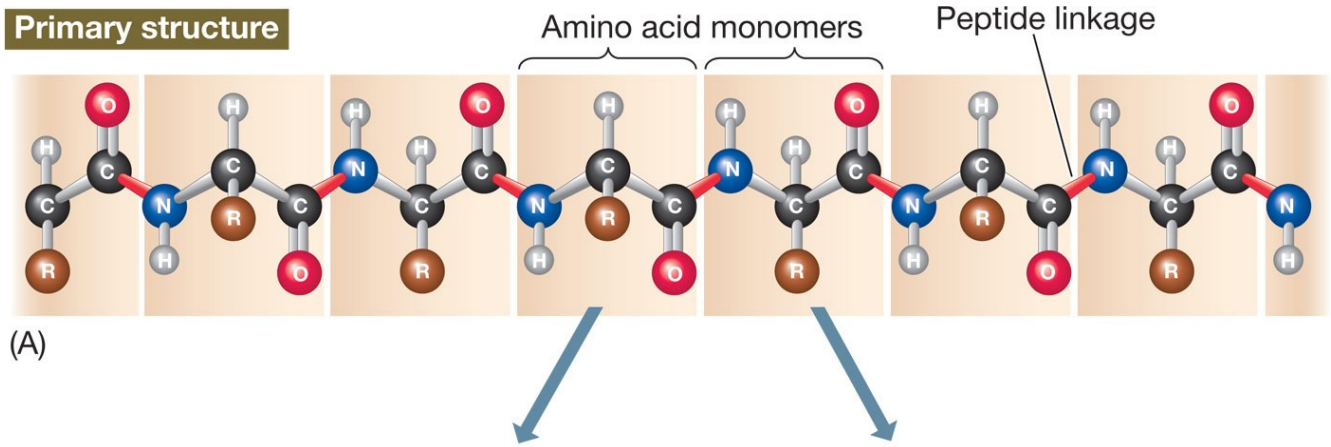
Primary sequence

There is directionality in polypeptide sequence. Aminoacids (aa) are listed from N terminus towards C terminus.

N-aa-aa-aa-aa-aa-C

Primary structure

- The **primary structure** of a protein is the sequence of amino acids.
- The sequence determines secondary and tertiary structure—how the protein is folded. Therefore, many copies of the same protein inside the cell have the same sequence.
- The number of different proteins that can be made from 20 amino acids is enormous!

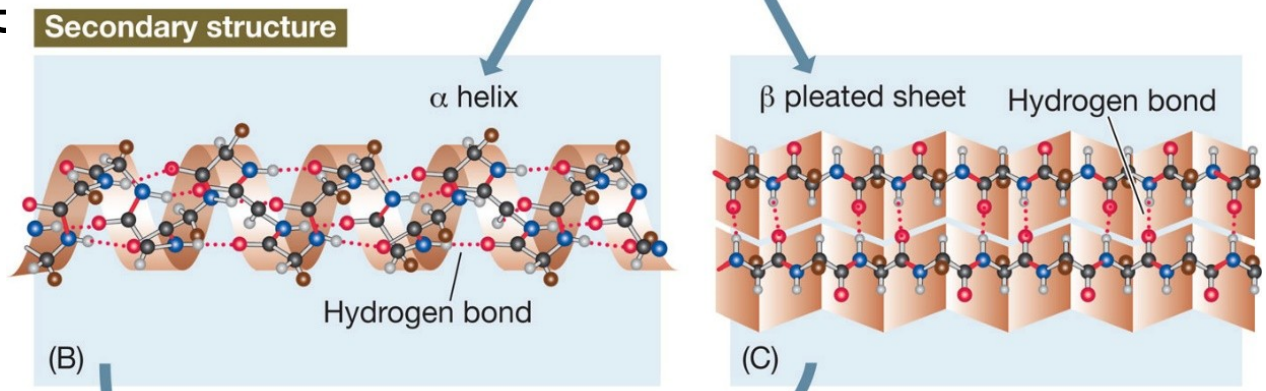


LIFE 10e, Figure 3.7 (Part 1)
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Secondary Structure

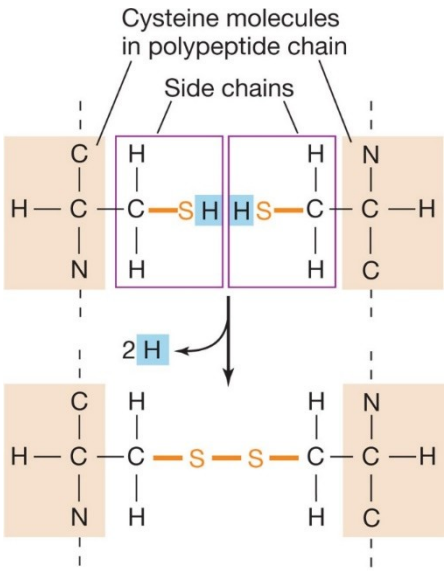
- **α helix**—right-handed coil resulting from hydrogen bonding between N–H groups on one amino acid and C=O groups on another.
- **β pleated sheet**—two or more polypeptide chains are aligned; hydrogen bonds form between the chains.

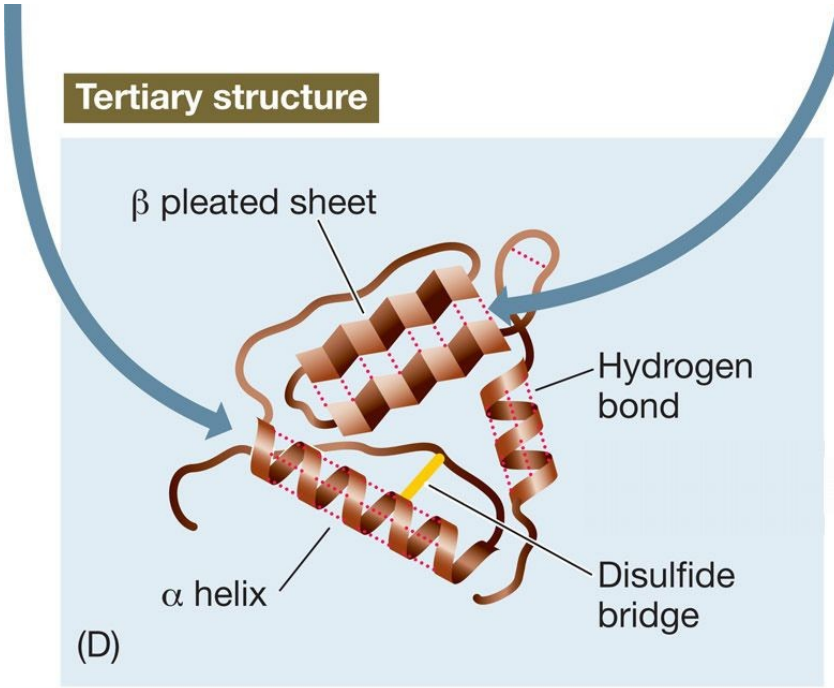
Hydrogen bonding between aminoacid side chains is essential for secondary structure of prote



Tertiary structure

- Bending and folding results in a macromolecule with specific three-dimensional shape.
- The outer surfaces present functional groups that can interact with other molecules.
- The terminal —SH group of cysteine can react with another cysteine side chain to form a **disulfide bridge**,).





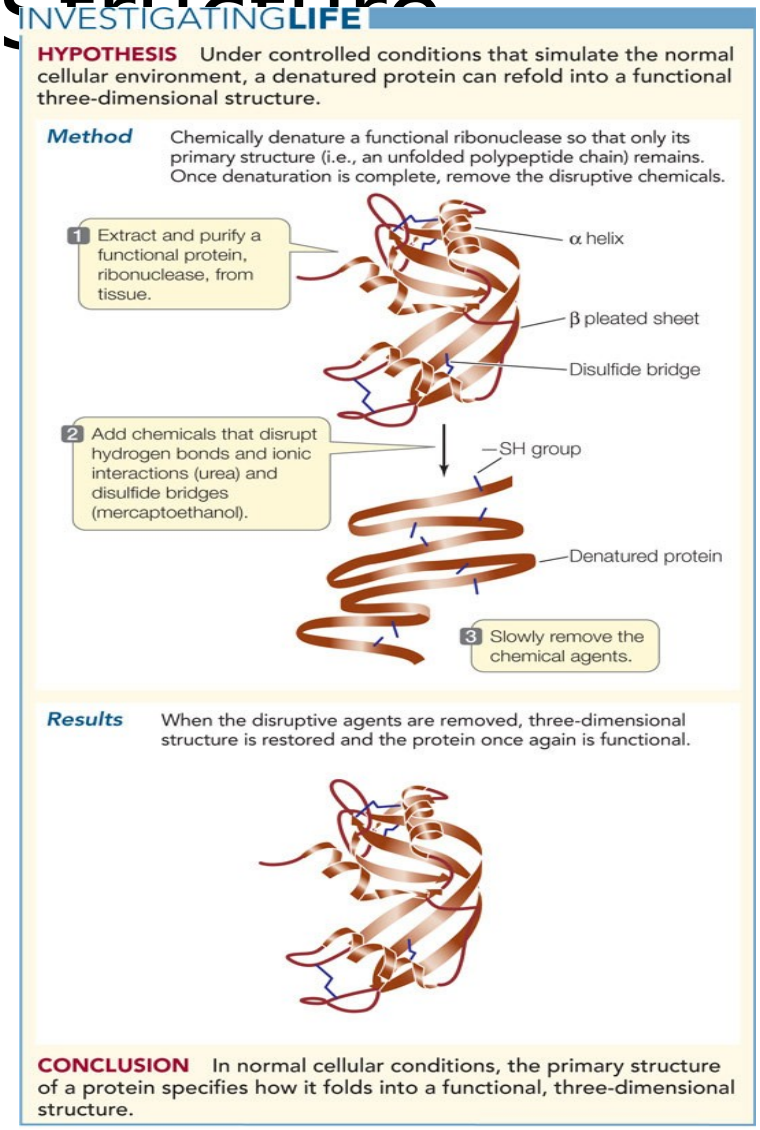
Tertiary structure

- Tertiary structure is determined by interactions between R-groups of aminoacids:
    - Disulfide bridges
    - Hydrogen bonds
    - Aggregation of hydrophobic side chains
    - Ionic attractions

Let’s have a look at real protein structures

- http://www.rcsb.org/pdb/explore/jmol.do?structureId=3ZOJ&bionumber=1&jmolMode=HTML5
- http://www.rcsb.org/pdb/explore/jmol.do?structureId=3EMN&bionumber=1&jmolMode=HTML5

Primary Structure Specifies Tertiary Structure

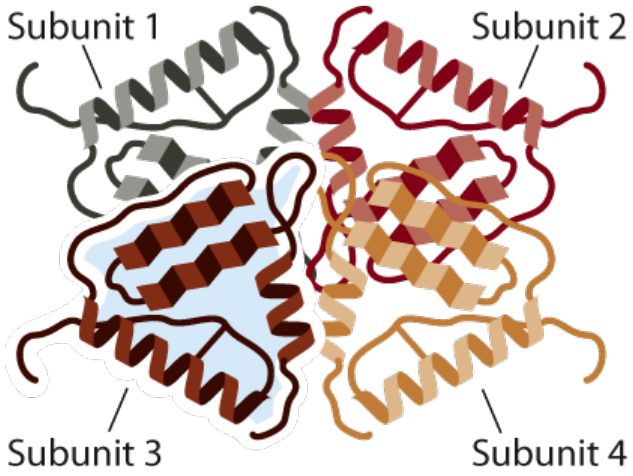


***LIFE 10e*, Figure 3.10**

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Quaternary Structure

- **Quaternary structure** results from the interaction of subunits (multiple polypeptides) by hydrophobic interactions, van der Waals forces, ionic attractions, and hydrogen bonds.
- Each subunit has its own unique tertiary structure.
- Quaternary Structure: Two or more polypeptides assemble to form larger protein molecules. The hypothetical molecule here is a tetramer, made up of four polypeptide subunits.

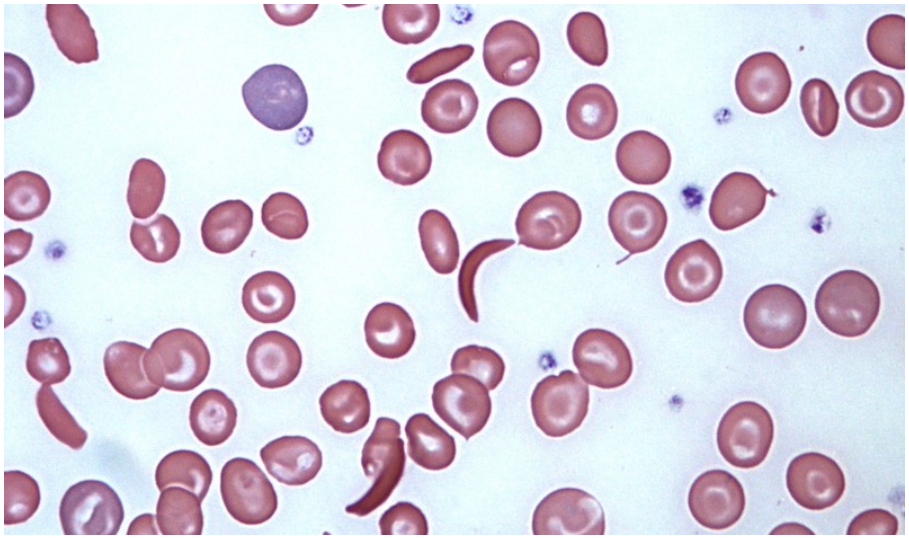


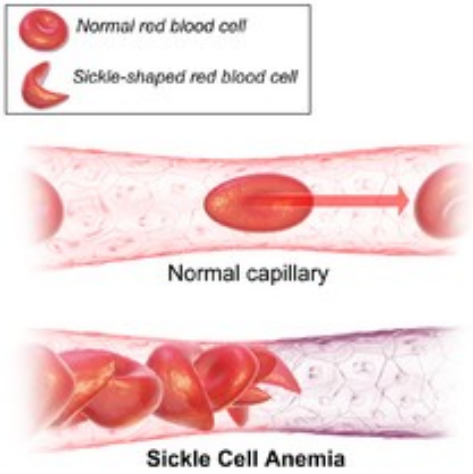
Hemoglobin

- https://www.rcsb.org/pdb/explore/jmol.do?structureId=1A3N&bionumber=1&jmolMode=HTML5

Sickle cell disease

- A single nucleotide point mutation in beta-hemoglobin (A to T) results in a substitution of Glutamate to Valine.
- http://www.rcsb.org/pdb/explore/jmol.do?structureId=2HBS&opt=3

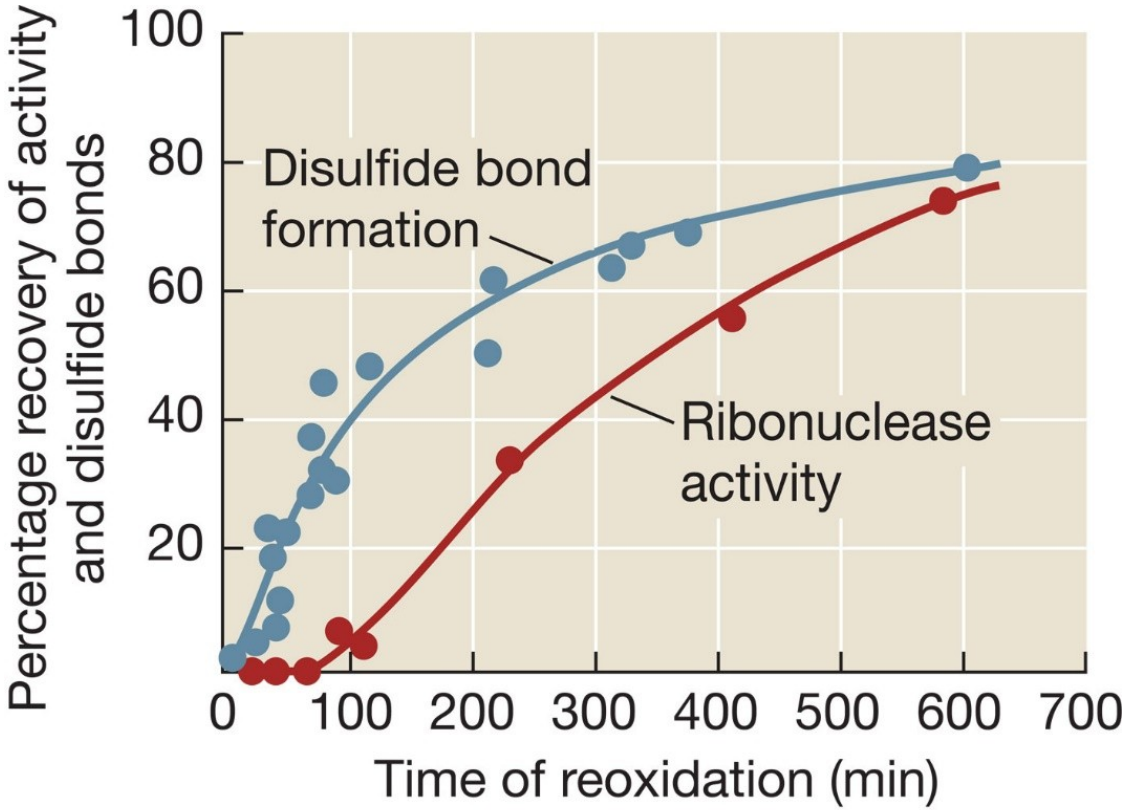




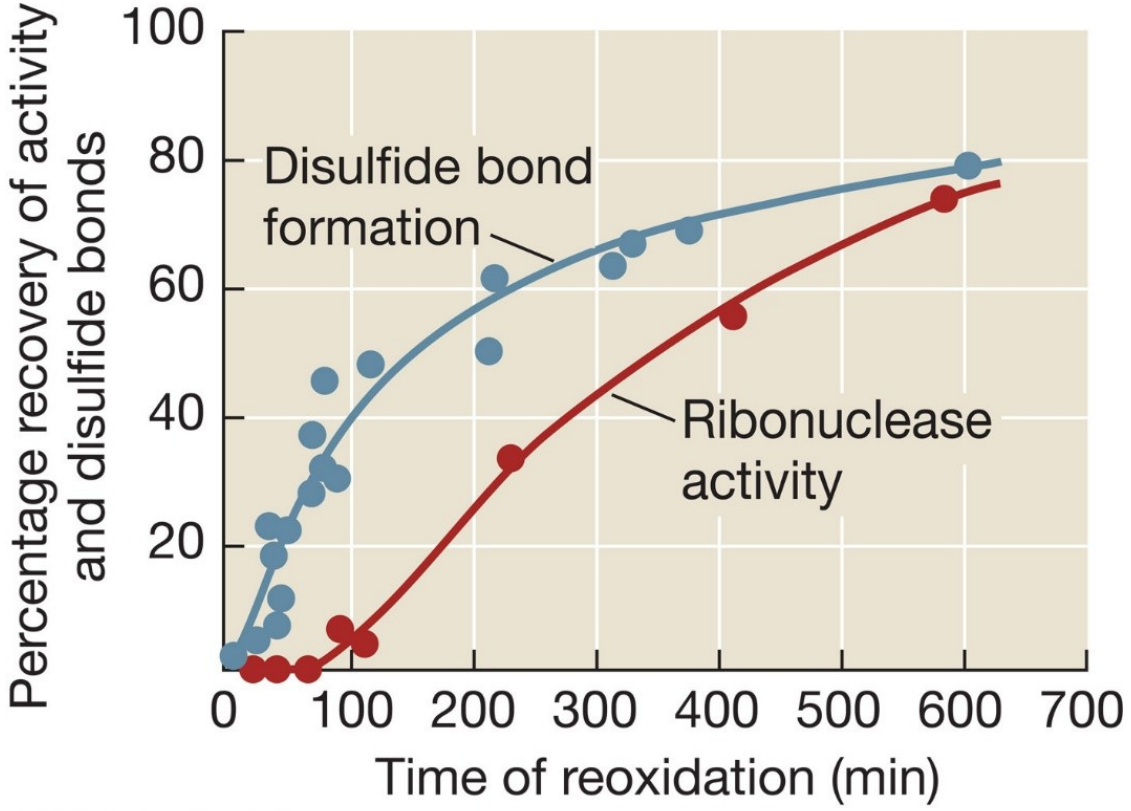
Protein Denaturation

- Conditions that affect secondary and tertiary structure:
  - High temperature
  - pH changes
  - High concentrations of polar molecules
  - Nonpolar substances
- If a protein is heated, secondary and tertiary structure break down; the protein is said to be **denatured**.
- When cooled, the protein returns to normal tertiary structure, demonstrating that the information to specify protein shape is in the primary structure. **Renaturation.** Not always possible.

The disulfide bonds were destroyed chemically, then allowed to reform. Protein function was assessed by enzyme activity.



LIFE 10e, Working with Data 3.1, Figure A
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LIFE 10e, Working with Data 3.1, Figure A
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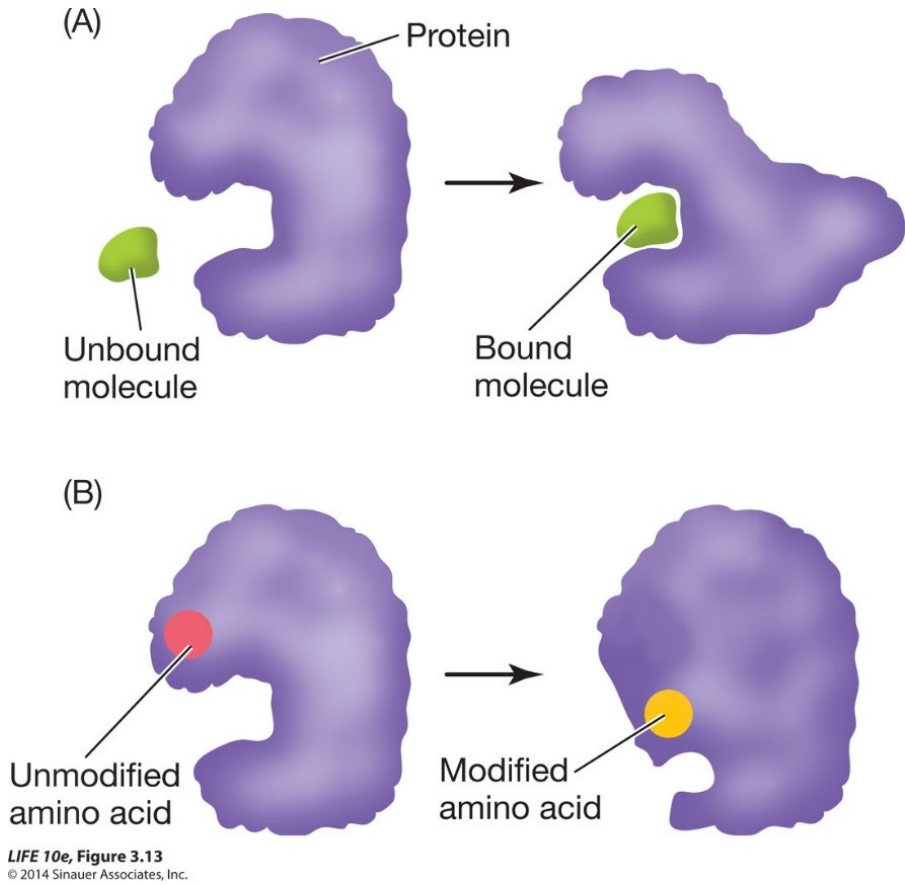
- At what time did disulfide bonds begin to form?
- At what time did enzyme activity begin to appear?
- Explain the difference between these times.

How do proteins interact with other molecules?

- Proteins bind noncovalently with specific molecules. Specificity is determined by:
- *Shape*—there must be a general “fit” between the 3-D shapes of the protein and the other molecule.
- *Chemistry*—R groups on the surface interact with other molecules via ionic, hydrophobic, or hydrogen bonds.

Protein shape can change

- Protein shape can change as a result of:
  - Interaction with other molecules— e.g., an enzyme changes shape when it comes into contact with a reactant.
  - Covalent modification— addition of a chemical group to an amino acid.



LIFE 10e, Figure 3.13
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Hemoglobin oxygenation

- http://pdb101.rcsb.org/motm/41

Nucleic acids

Nucleic Acids

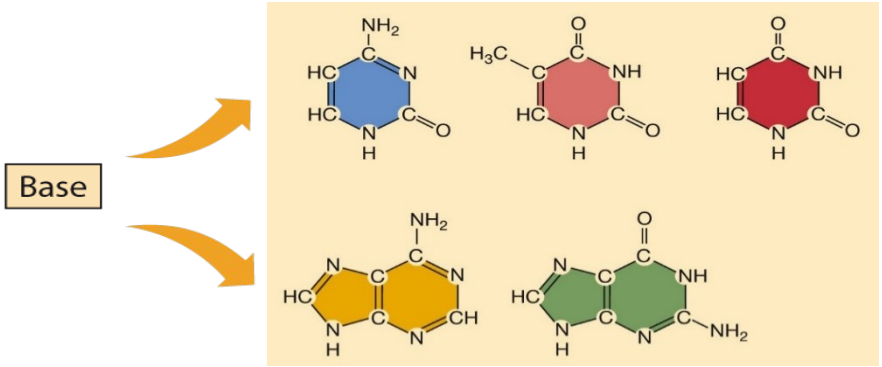
- **Nucleotides** are the monomers that make up nucleic acids.
- Nucleotides consist of a pentose sugar, a phosphate group, and a nitrogen-containing **base**.
- A **nucleoside** consists only of a pentose sugar and a nitrogenous base.

Nucleotides are composed of three components:

1. A nitrogen-containing base

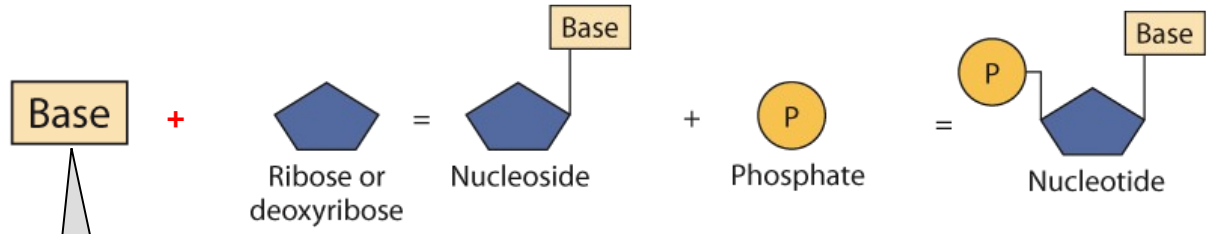
   The base may be either a pyrimidine or a purine.

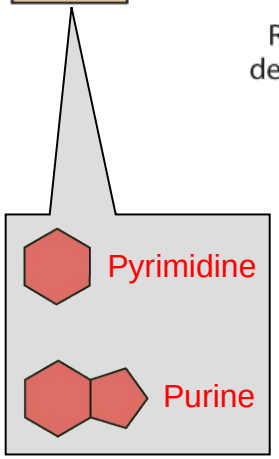
Pyrimidines have a single ring.



Purines have two fused rings.

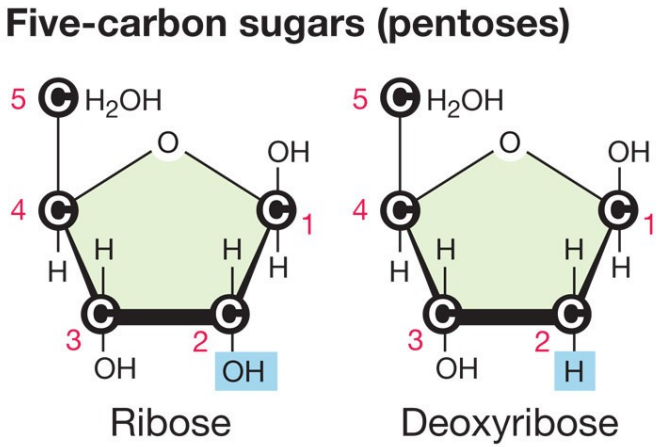
2. A pentose sugar (ribose or deoxyribose)
3. A phosphate group





Nucleic acid structure

- RNA contains the sugar **ribose**.
- DNA contains **deoxyribose**.



4.1 What Are the Chemical Structures and Functions of Nucleic Acids?

- DNA bases:
  - **Adenine** (**A**)
  - **Cytosine** (**C**)
  - **Guanine** (**G**)
  - **Thymine** (**T**)
- RNA has **uracil** (**U**) instead of t

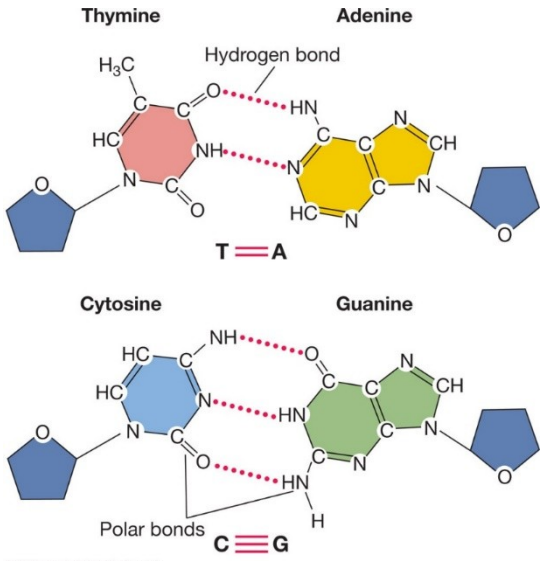
TABLE **4.1**
Distinguishing RNA from DNA

Nucleic Acid	Sugar	Bases	Name of Nucleoside	Strands
RNA	Ribose	Adenine	Adenosine	Single
		Cytosine	Cytidine	
		Guanine	Guanosine	
		Uracil	Uridine	
DNA	Deoxyribose	Adenine	Deoxyadenosine	Double
		Cytosine	Deoxycytidine	
		Guanine	Deoxyguanosine	
		Thymine	Deoxythymidine	

LIFE 10e, Table 4.1

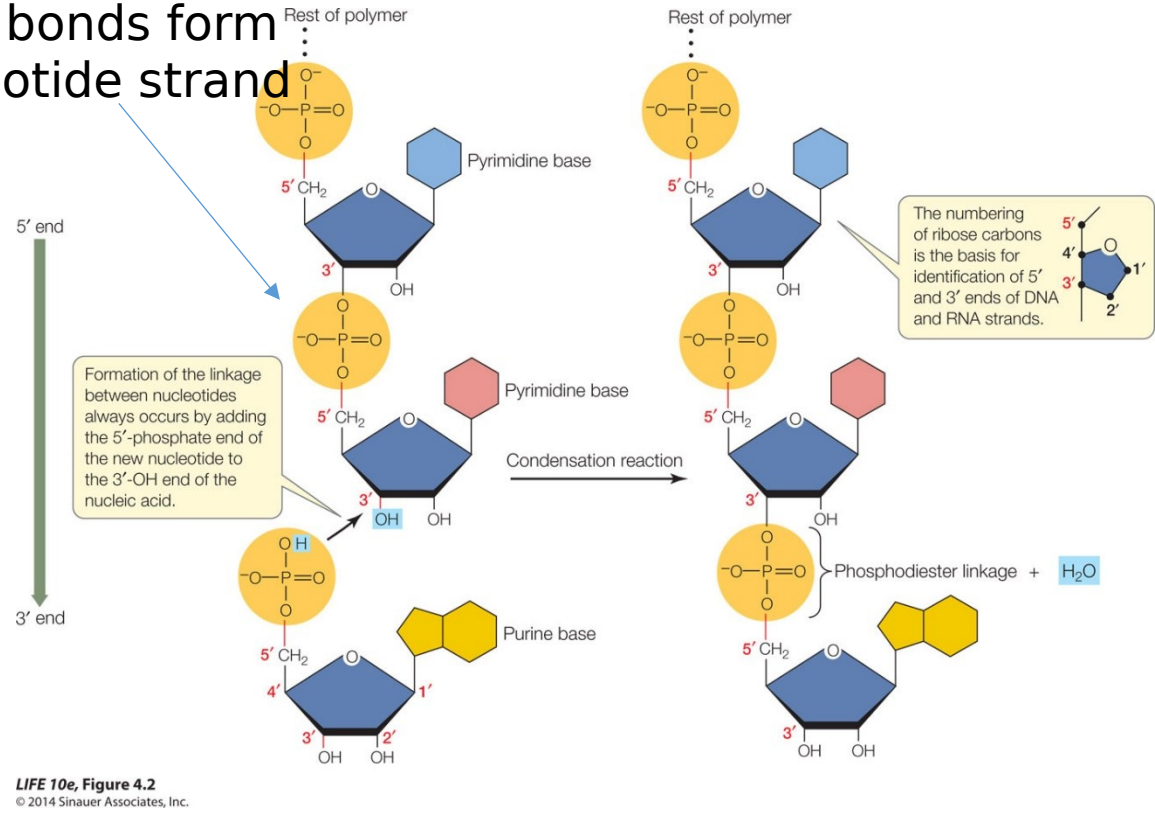
How to build a nucleic acid from nucleotides?

- **Complementary base pairing**: purines pair with pyrimidines by **hydrogen bonds**.



How to build a nucleic acid from nucleotides?

Covalent bonds form the nucleotide strand



- There is directionality to this strand: 5’ phosphate to 3’OH group (5’-3’).
- All DNA and RNA sequences are written in 5’->3’ direction.

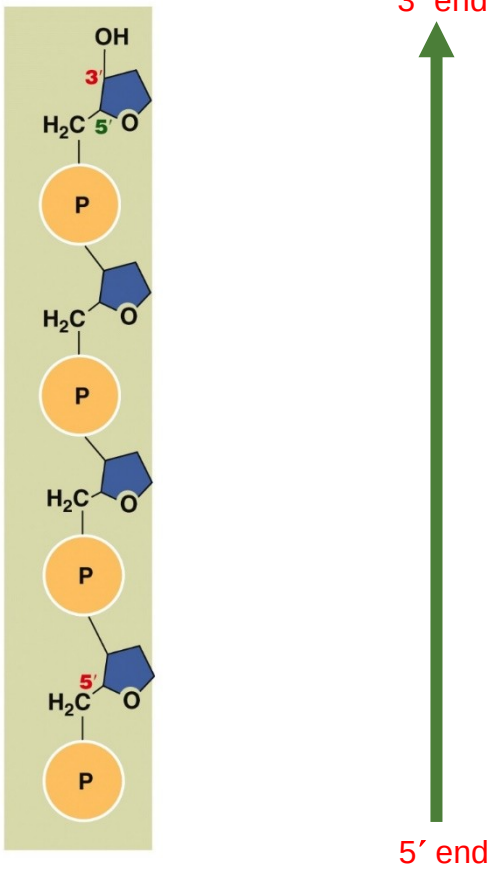
www.rcsb.org/pdb/explore/jmol.do?structureId=1BNA&bionumber=1&jmolMode=
db101.rcsb.org/motm/23

RNA

RNA (single-stranded)

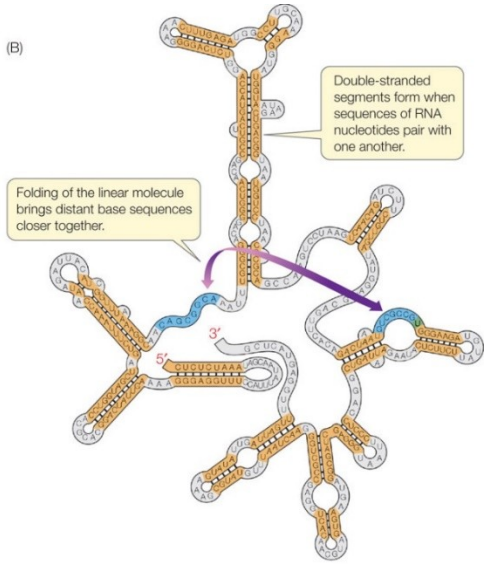
In RNA, the bases are attached to the ribose.

The bases are adenine (A) and guanine (G), the purines; and cytosine (C) and uracil (U), the pyrimidines.



RNA

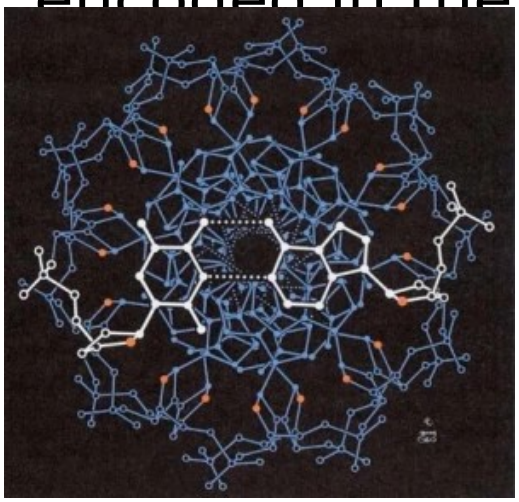
- RNA is single-stranded, but base pairing occurs between different regions of the molecule. Therefore, the 3D structure of RNA is not a single-strand.
- Base pairing determines the three-dimensional shape of some RNA molecules.
- Complementary base pairing can also take place between RNA and DNA.



http://www.rcsb.org/pdb/explore/jmol.do?structureId=4GXY&bionumber=1&jmolMode=HTML5

DNA

- The two strands of a DNA molecule form a **double helix**.
- All DNA molecules have the same structure (unlike proteins); diversity lies in the sequence of base pairs.
- DNA is an informationa ation is encoded in the sequen



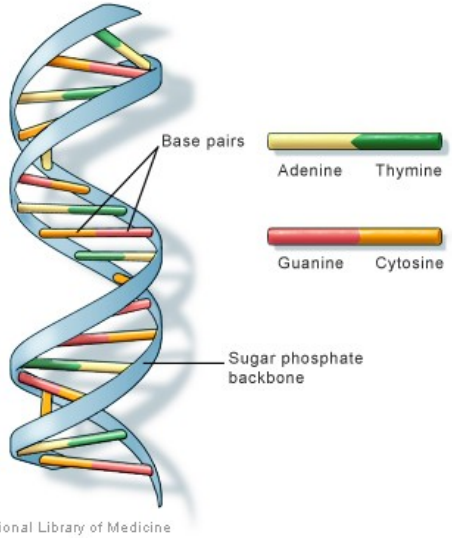
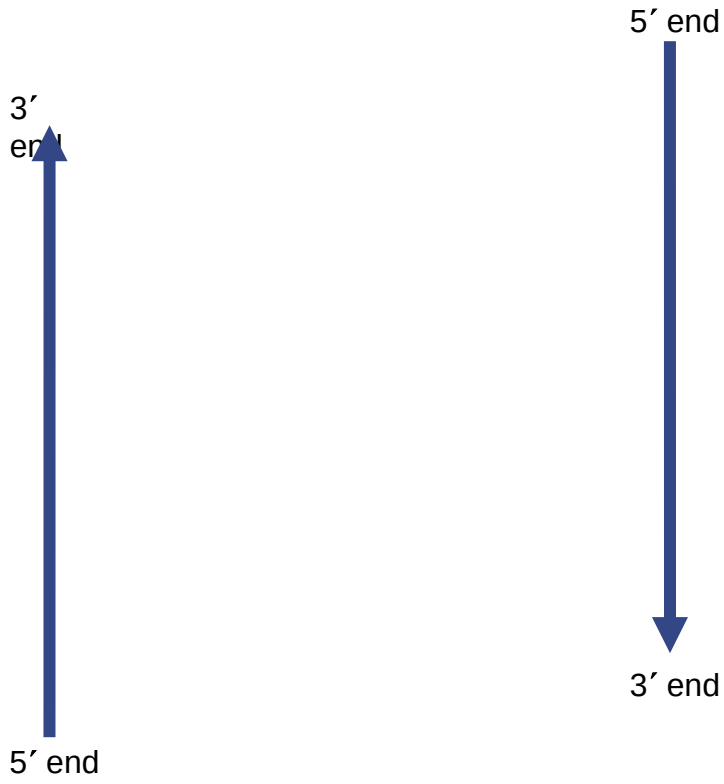


Figure 4.4 DNA

DNA (double-stranded)

In DNA, the bases are attached to deoxyribose. The bases are adenine (A), guanine (G), cytosine (C), and thymine (T) (thymine is found instead of uracil). DNA is double-stranded.

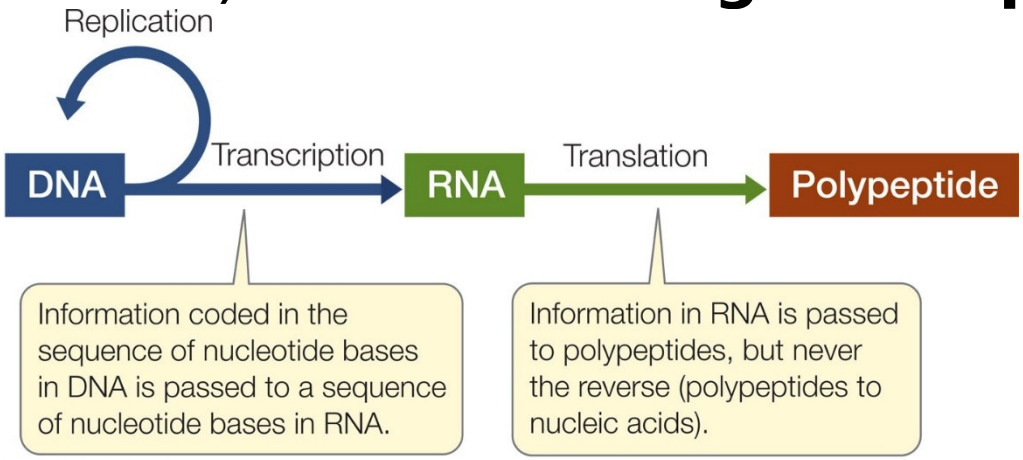
Hydrogen bonds between purines and pyrimidines hold the two strands of DNA together.



Two strands of DNA are antiparallel to each other: One strand goes from 5’-3’ direction, the other 3’-5’ direction.

Central Dogma

- DNA transmits information in two ways:
- DNA can reproduce itself (**replication**).
- DNA sequences can be copied into RNA (**transcription**). The RNA can specify a sequence of amino acids in a polypeptide (**translation**). This is called **gene expression**.



LIFE 10e, In-Text Art, Ch. 4, p. 65
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Functions of Nucleic acids

- DNA carries hereditary information between generations.
- RNA is the middle-man for gene expression.
- Determining the sequence of bases helps reveal evolutionary relationships.
- Other roles for nucleotides:
  - ATP—energy transducer in biochemical reactions
  - GTP—energy source in protein synthesis
  - cAMP—essential to the action of hormones and transmission of information in the nervous system